

Vihanga Rathnayake

Final-Year Information Systems Undergraduate | Business Analysis | Systems Analysis | Digital Transformation
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[Linkedin](#) | [GitHub](#) | [Portfolio](#)

Profile

Motivated Information Systems undergraduate with experience in business analysis, requirements engineering, systems analysis, business process reengineering, database design, enterprise systems, and digital transformation. Published researcher and project developer currently contributing to a government Digital Case Flow Transformation initiative involving stakeholder analysis, BPMN and UML modelling, system requirements, cloud architecture, and prototype design. Skilled in translating complex operational requirements into structured processes and practical technology solutions. Seeking opportunities in Business Analysis, Systems Analysis, ERP, Solutions Consulting, or Digital Transformation.

Education

Bachelor of Computing (Hons.) in Information Systems June 2023 - Present
University of Sri Jayewardenepura | Final Year | CGPA after 6th Semester: 3.21/4.00

Pearson Assured Diploma in Information Technology Completed 2022
ESOFT Metro College, Monaragala | Distinction Class

Vadya Visharad in Tabla Playing Completed 2017
Bhathkhade Sangeeth Vidyapith, India | Distinction Class

Selected Projects

Government Digital Case Flow Management System - National Authority for the Protection of Victims of Crime and Witnesses , Ministry of Justice and National Integration July 2026 - Present
Technologies: Figma, BPMN, UML, Bizagi Modeler, Requirements Engineering, RBAC, UI/UX Prototyping
Contributed to this Digital transformation initiative by conducting stakeholder interviews, requirements elicitation, and business process analysis across five priority authority divisions. Mapped existing complaint and case-management workflows, identified operational gaps, and designed improved As-Is and To-Be processes. Translated stakeholder requirements into use cases, functional and non-functional requirements, security controls, role matrices, data definitions, system architecture, and an interactive role-based prototype with downloadable case-summary reporting.

[\[SOURCE\]](#)

An Extensible ML-Driven Platform for Automated Report Generation in Geotechnical Engineering - Final Year Research Project March 2026 - Present
Technologies: Python, FastAPI, scikit-learn, Random Forest, SVM, TensorFlow, pandas, OpenCV, React.js, PostgreSQL, REST APIs, ReportLab, python-docx
Designing and evaluating an extensible ML-driven platform for automated report generation in geotechnical engineering using the Design Science Research methodology. The initial implementation focuses on Low Strain Pile Integrity Testing (PIT), integrating document parsing, waveform extraction, signal-feature engineering, hybrid rule-based and machine-learning classification, explainable results, and ASTM D5882-aligned report generation through a React-based application. The data pipeline uses more than 22,000 Sri Lankan bore pile records, while the modular architecture is designed to support future integration of PDA, CSL, and other geotechnical testing methods. [\[SOURCE\]](#)

PrimeHeal - Smart Healthcare Support System August 2025 - June 2026
Technologies: MySQL, React.js, Node.js, Express.js, JWT, Tailwind CSS
Served as Database Administrator for a group platform supporting patient records, appointments, and role-specific dashboards. Designed normalized relational schemas, queries, indexing strategies, and integrity controls for secure and scalable data operations. [\[SOURCE\]](#)

ERP Implementation and official Website - PiLETesT Consultants (Pvt) Ltd July 2026 - Present
Technologies: HTML, CSS, JS, TypeScript, Astro, Odoo 18, BPMN, Business Process Analysis, Requirements Engineering, ISO 9001
Assessed project management, procurement, inventory, equipment, finance, and HR processes; developed requirements, BPMN models, and an Odoo implementation roadmap to improve workflow integration, quality compliance, automation, and management decision-making.

[\[SOURCE\]](#)

Pawfect - AI-Powered Pet Adoption and Care Mobile Application 2026
Technologies: Android, Java, TensorFlow Lite, Firebase, AI Recommendation, Mobile UI/UX
Developed a mobile application connecting adopters with registered animal shelters and recommending pets based on lifestyle preferences. Contributed secure user journeys, shelter workflows, and post-adoption care tools.

[\[SOURCE\]](#)

Research Publications

Digital Transformation in Geotechnical Engineering: A Systematic Review of Mobile Systems and AI-Driven Report Generation

1st Jayewardenepura Computing Symposium (JCoS), University of Sri Jayewardenepura, 2026 - Co-author and Presenter

Reviewed mobile geotechnical systems, AI-assisted report generation, offline-first designs, retrieval-augmented generation, and organizational barriers to digital transformation.

Technical Skills

- **Business Analysis:** Requirements elicitation, stakeholder interviews, gap analysis, process mapping, functional specifications, use cases, and traceability.
- **Process and System Modelling:** BPMN, UML, As-Is/To-Be analysis, BPR, data dictionaries, role matrices, and role-based access control.
- **Programming and Web:** Python, Java, JavaScript, TypeScript, C#, C++, React.js, Node.js, Express.js, FastAPI, and REST APIs.
- **Data and Machine Learning:** MySQL, PostgreSQL, MongoDB, SQL, schema design, pandas, scikit-learn, TensorFlow, and model evaluation.
- **Enterprise, Design, and Delivery:** Odoo 18, Figma, Git, GitHub, AWS fundamentals, GitHub Actions, and CI/CD.

Non-Technical Skills

- **Problem Solving:** Strong analytical and problem-solving skills with attention to detail
- **Curiosity:** Passionate about exploring emerging technologies and continuously expanding technical knowledge
- **Teamwork:** Effective collaboration in cross-functional and team environments
- **Leadership:** Passionate about lead a team for achieving a common goal
- **Teaching Mentoring:** Experience mentoring and teaching technical

Certifications

- **Networking Basics** - Cisco [\[SOURCE\]](#)
- **Google Analytics** - Google [\[SOURCE\]](#)
- **Web Designing** - Open Learning Platform - UoM [\[SOURCE\]](#)

Selected Leaderships

President, Union Literature Subcommittee, University of Sri Jayewardenepura *May 2025 - March 2026*
Organized and coordinated cultural and literal initiatives within the university community such as Himidiriya , Wedikawen Mahapolowata

Secretary, Flair Club, University of Sri Jayewardenepura *March 2025 - June 2026*
Managed executive documentation, reporting, stakeholder coordination, and project supervision for university-wide initiatives in cooperation with Career Guidance Unit of the university

Co-Founder and Secretary, Association of Computing Students *February 2024 - June 2025*
Coordinated executive operations, stakeholder engagement, and academic-development programs.

Founder and Chairperson, Project EduACS *March 2024 - May 2025*
Initiated and led islandwide G.C.E. A/L IT and O/L Mathematics outreach seminars, coordinating speakers, partner stakeholders, and project teams to enhance dignity of Faculty of Computing USJ

Faculty Board Representative, Faculty of Computing *January 2024 - March 2025*
Represented student perspectives at faculty-level meetings

Achievements

- **Logistics and Facilitation Excellence Award**, Flair Club (June 2025)
- **Member of the Month**, G17 Global (July 2025)
- **Environmental Pioneer Presidential Award**, Central Environmental Authority (November 2017)

References

Prof. Prasad M. Jayaweera
The Dean
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Mrs. Chathurangani Thennakoon
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